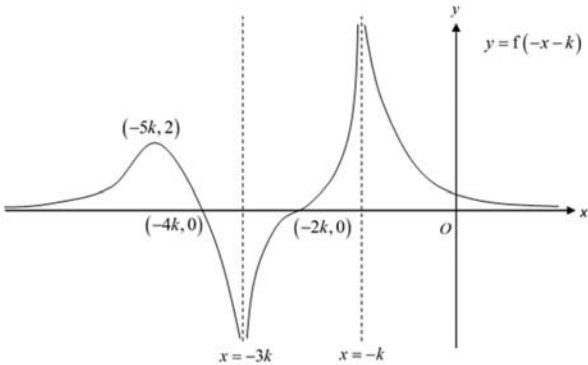
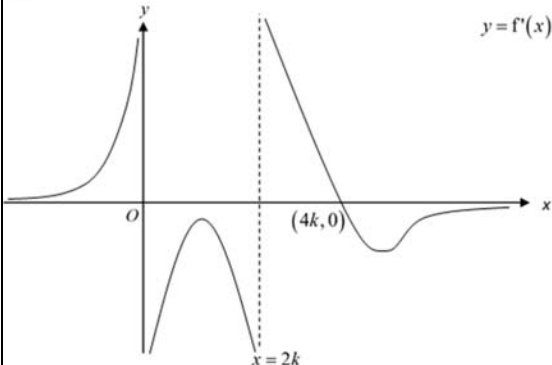
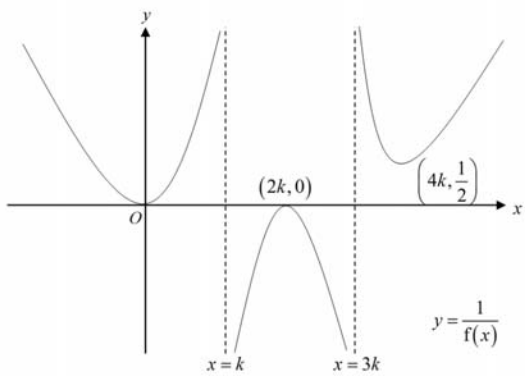
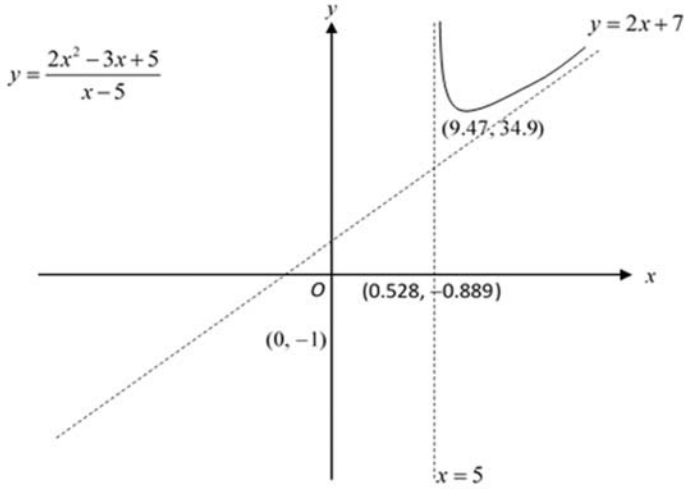
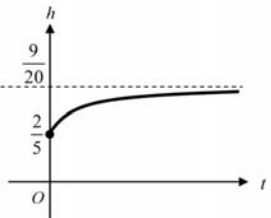


ANNEX B

CJC H2 Math JC2 Preliminary Examination Paper 1

QN	Topic Set	Answers
1	Graphs and Transformation	(ii) $k < -\frac{1}{4}$
2	Application of Integration	$\pi(2\ln 2 + 2)$
3	Sigma Notation and Method of Difference	(i) $\frac{1}{(r+1)!} - \frac{1}{(r+2)!}$ (ii) $\frac{1}{3} \left[\frac{1}{2} - \frac{1}{(n+2)!} \right]$ (iii) $\frac{1}{3}$
4	AP and GP	(i) \$197.10 (iii) August 2017
5	Graphs and Transformation	(i)  (ii)  (iii)

		
6	Differentiation & Applications	(ii) $7 + 4\sqrt{3}$
7	Differentiation & Applications	(i) $-\frac{t(t+1)^2}{2}$ (ii) $y = -\frac{1}{6}x + \frac{17}{3}$ or $6y + x = 34$ (iii) $t = -3$ (given) or -0.915 or 2.91
8	Graphs and Transformation	(i) $p = 2, q = 7, r = 40$ (ii)  (iii) 2
9	Maclaurin series	(a) $p = 2 - \ln 4$ (b)(i) $\frac{\pi}{4} + \frac{1}{2}ax - \frac{1}{4}a^2x^2 + \frac{1}{12}a^3x^3 + \dots$ (ii) $\frac{1}{2} - 2x + 4x^2$

10	Differential Equations	(ii) $h = \frac{1}{20} \left(9 - e^{-\frac{1}{20}t} \right)$ (iv) 
11	Vectors	(iv) $(1, -38, -57)$ (v) $l_m : x = 1, y - 2 = \frac{z - 3}{8}$



Q1. Transformations, Conics and Inequalities		
Assessment Objectives	Solution	Examiner's Feedback
Determine the transformations on the graph of $y = f(x)$ as represented by $y = f(x) + a$ and $ay = f(x)$.	(i) $(y - a)^2 = ax$ $\downarrow A$ $(y - a)^2 = a(x - 2)$ $\downarrow B$ $(3y - a)^2 = a(x - 2)$ Since resulting curve passes through point $\left(2, \frac{4}{3}\right)$, $(4 - a)^2 = a(2 - 2)$ $(4 - a)^2 = 0$ $a = 4$ (shown)	Most candidates were able to answer this part correctly. Some forgot that scaling to a variable is achieved by dividing the variable by the scaling factor.
Applying the concept no real roots $\Rightarrow b^2 - 4ac < 0$	(ii) Method ①: Parabola: $(y - 4)^2 = 4x$ — ① Line: $y = kx$ — ② Substitute ② into ①: $(kx - 4)^2 = 4x$ $k^2x^2 - 8kx + 16 = 4x$ $k^2x^2 + (-8k - 4)x + 16 = 0$ For the line not to meet the parabola, $b^2 - 4ac < 0$	Many presented satisfactory answers. Some students failed to link the intersection of linear/quadratic curves to solving simultaneous and subsequently quadratic equations, and that the number of common points can be inferred from the sign of the determinant. Some also had algebraic slips when handling inequalities. They need to practise more.

$$(-8k - 4)^2 - 4k^2(16) < 0$$

$$64k^2 + 64k + 16 - 64k^2 < 0$$

$$64k + 16 < 0$$

$$k < -\frac{1}{4}$$

Method ②:

Parabola: $(y - 4)^2 = 4x$ — ①

Line: $y = kx \Rightarrow x = \frac{k}{y}$ — ②

Substitute ② into ①:

$$(y - 4)^2 = \frac{4y}{k}$$

$$ky^2 - 8ky + 16k = 4y$$

$$ky^2 + (-8k - 4)y + 16k = 0$$

For the line not to meet the parabola, $b^2 - 4ac < 0$

$$(-8k - 4)^2 - 4k(16k) < 0$$

$$64k^2 + 64k + 16 - 64k^2 < 0$$

$$64k + 16 < 0$$

$$k < -\frac{1}{4}$$

Q2. Definite Integral		
Assessment Objectives	Solution	Examiner's Feedback
Find the volume of solid formed by revolution. Perform integration by a given substitution.	Method ①: Volume $= \pi \int_9^{16} \left(\frac{1}{\sqrt{x}-2} \right)^2 dx$ $= \pi \int_3^4 \left(\frac{1}{t-2} \right)^2 (2t) dt$ $= \pi \int_3^4 \frac{2t}{t^2-4t+4} dt$ $= \pi \int_3^4 \frac{2t-4}{t^2-4t+4} + \frac{4}{(t-2)^2} dt$ $= \pi \left[\ln t^2-4t+4 \right]_3^4 + \pi \int_3^4 4(t-2)^{-2} dt$ $= \pi \left[\ln t^2-4t+4 + 4 \frac{(t-2)^{-1}}{-1} \right]_3^4$ $= \pi \left[\ln t^2-4t+4 - \frac{4}{(t-2)} \right]_3^4$ $= \pi [(\ln 4 - 2) - (\ln 1 - 4)]$ $= \pi (\ln 4 + 2) \text{ units}^3$ <u>Substitution:</u> $t = \sqrt{x}$ $t^2 = x$ $2t = \frac{dx}{dt}$ When $x = 9, t = 3$ When $x = 16, t = 4$	Most candidates were able to setup the correct integral for the volume of revolution. However, many failed make the correct substitution of dx by $\frac{dx}{dt} dt$ and thus $2t dt$. Another group of students forgot to change the upper and lower limits to the respective values of t when the variable was changed. Many students were also stuck at the integration of $\frac{4}{(t-2)^2}$ as it is not very easy for those who don't practise much to identify the fraction as a power function of power -2.

Method ②:

Volume

$$= \pi \int_9^{16} \left(\frac{1}{\sqrt{x}-2} \right)^2 dx$$

$$= \pi \int_3^4 \left(\frac{1}{t-2} \right)^2 (2t) dt$$

$$= \pi \int_3^4 \frac{2t}{(t-2)^2} dt$$

$$= \pi \int_3^4 \frac{2}{(t-2)} + \frac{4}{(t-2)^2} dt$$

$$= \pi \left[2 \ln |t-2| \right]_3^4 + \pi \int_3^4 4(t-2)^{-2} dt$$

$$= \pi \left[2 \ln |t-2| + 4 \frac{(t-2)^{-1}}{-1} \right]_3^4$$

$$= \pi \left[2 \ln |t-2| - \frac{4}{(t-2)} \right]_3^4$$

$$= \pi \left[(2 \ln 2 - 2) - (2 \ln 1 - 4) \right]$$

$$= \pi (2 \ln 2 + 2) \text{ units}^3$$

Substitution:

$$t = \sqrt{x}$$

$$t^2 = x$$

$$2t = \frac{dx}{dt}$$

$$\text{When } x=9, t=3$$

$$\text{When } x=16, t=4$$

Q3. Sigma Notation		
Assessment Objectives	Solution	Examiner's Feedback
Apply concept of factorial	(i) Method ①: $\frac{r+1}{(r+2)!} = \frac{A}{(r+1)!} + \frac{B}{(r+2)!}$ $r+1 = \frac{A(r+2)!}{(r+1)!} + \frac{B(r+2)!}{(r+2)!}$ $r+1 = A(r+2) + B$ When $r = -1$, $A+B=0$ — ① When $r = 0$, $2A+B=1$ — ② Solving, $A=1$ and $B=-1$ $\therefore \frac{1}{(r+2)r!} = \frac{1}{(r+1)!} - \frac{1}{(r+2)!}$ Method ②: $\frac{r+1}{(r+2)!} = \frac{A}{(r+1)!} + \frac{B}{(r+2)!}$ When $r=1$, $\frac{A}{2} + \frac{B}{6} = \frac{2}{6}$ $3A+B=2$ — ① When $r=0$, $A + \frac{B}{2} = \frac{1}{2}$ $2A+B=1$ — ② Solving, $A=1$ and $B=-1$ $\therefore \frac{1}{(r+2)r!} = \frac{1}{(r+1)!} - \frac{1}{(r+2)!}$	Most students were able to get the values of A and B correctly. There were a variety of methods used to get the correct answers.

Apply summation of series by the method of differences.	(ii) $\sum_{r=1}^n \frac{r+1}{3(r+2)!} = \frac{1}{3} \sum_{r=1}^n \frac{r+1}{(r+2)!}$ $= \frac{1}{3} \sum_{r=1}^n \left[\frac{1}{(r+1)!} - \frac{1}{(r+2)!} \right]$ $= \frac{1}{3} \left[\frac{1}{2!} - \frac{1}{3!} \right. \\ + \frac{1}{3!} - \frac{1}{4!} \\ + \dots \dots \\ + \frac{1}{n!} - \frac{1}{(n+1)!} \\ + \frac{1}{(n+1)!} - \frac{1}{(n+2)!} \left. \right]$ $= \frac{1}{3} \left[\frac{1}{2} - \frac{1}{(n+2)!} \right]$	Most students were able to get this part correct.
Understand convergence of a series and the sum to infinity.	(iii) From (ii), $\sum_{r=1}^n \frac{r+1}{3(r+2)!} = \frac{1}{3} \left[\frac{1}{2} - \frac{1}{(n+2)!} \right]$ As $n \rightarrow \infty$, $\frac{1}{(n+2)!} \rightarrow 0$, thus $\sum_{r=1}^n \frac{r+1}{3(r+2)!} \rightarrow \frac{1}{6}$ $\sum_{r=0}^{\infty} \frac{r+1}{3(r+2)!} = \frac{1}{3(2)(1)} + \sum_{r=1}^{\infty} \frac{r+1}{3(r+2)!}$ $= \frac{1}{6} + \frac{1}{6}$ $= \frac{1}{3}$	Most students were able to get the sum to infinity correct but failed to realized that the starting value of r had change.

Q4. Geometric Progression

Assessment Objectives	Solution					Examiner's Feedback
Determine sum of a finite geometric series	(i)			Amount of \$ Kumar has @ the ...		
				Beginning	Middle	End
		Jan 2017	1	100	$0.7(100)$	$0.7(100)+20$
		Feb 2017	2	$100+0.7(100)+20$	$0.7[100+0.7(100)+20]$	$0.7(100)+0.7^2(100)+0.7(20)+20$
		Mar 2017	3	$100+0.7(100)+0.7^2(100)+0.7(20)+20$	$0.7[100+0.7(100)+0.7^2(100)+0.7(20)+20]$	$0.7(100)+0.7^2(100)+0.7^3(100)+0.7^2(20)+0.7(20)+20$
				...	...	...
		n			$0.7(100)+0.7^2(100)+0.7^3(100)+\dots+0.7^n(100)+0.7^{n-1}(20)+\dots+0.7^2(20)+0.7(20)+20$	
	Amount of money Kumar has at the end of March 2015 $= \$[0.7(100)+0.7^2(100)+0.7^3(100)+0.7^2(20)+0.7(20)+20]$ $= \$197.10$					Most students were able to get the value correct.
	(ii)	Amount of money Kumar has at the end of n months $= 0.7(100)+0.7^2(100)+0.7^3(100)+\dots+0.7^n(100)+0.7^{n-1}(20)+\dots+0.7^2(20)+0.7(20)+20$ $= 100(0.7+0.7^2+\dots+0.7^n)+20(1+0.7+0.7^2+\dots+0.7^{n-1})$ $= 100\left[\frac{0.7(1-0.7^n)}{1-0.7}\right]+20\left[\frac{1-0.7^n}{1-0.7}\right]$ $= \frac{700}{3}(1-0.7^n)+\frac{200}{3}(1-0.7^n)$ $= 300(1-0.7^n)$ (shown)				

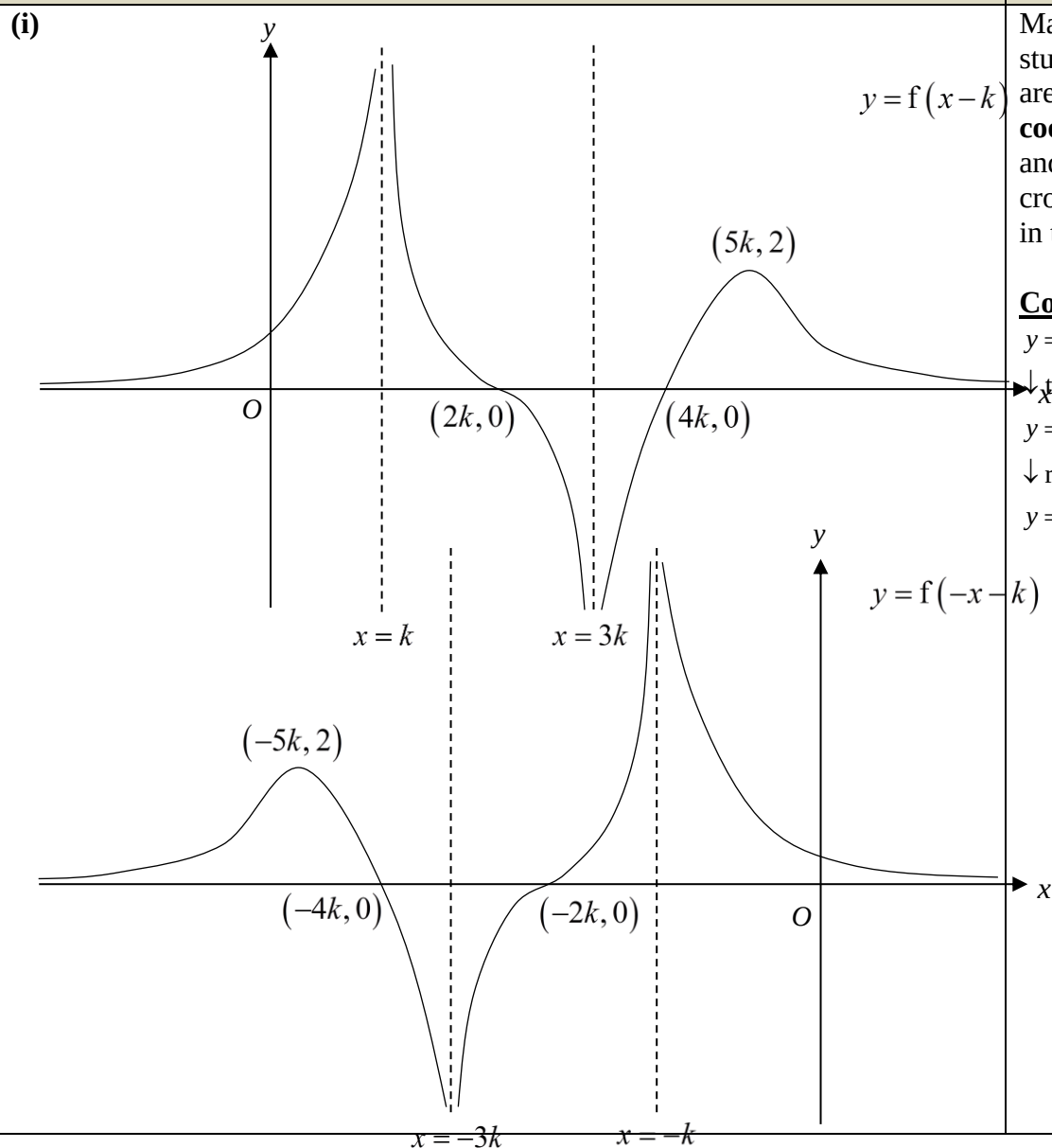
Solve inequality	(iii) $300(1 - 0.7^n) \geq 280$ $1 - 0.7^n \geq \frac{14}{15}$ $0.7^n \leq \frac{1}{15}$ $n \geq 7.59$ Kumar will first be able to purchase the gift for his mother at the 8th month. (or August 2017)	Quite badly done by students who did not use the GC table method with many students not realizing that $\ln 0.7 < 0$ and hence there is a need to change the inequality sign when dividing by $\ln 0.7$ on both sides of the inequalities.
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Q5. Transformations

Assessment Objectives

Apply effect of transformation on the graph of $y = f(x)$ as represented by $y = f(-x - k)$.

Solution



Examiner's Feedback

Many good answers. However, students should take note that they are supposed to state the **coordinates** of any turning point(s) and any points where the curve crosses the x- and y-axes as given in the question.

Correct sequence:

$$y = f(x)$$

↓ translate k units in the +ve x -direction

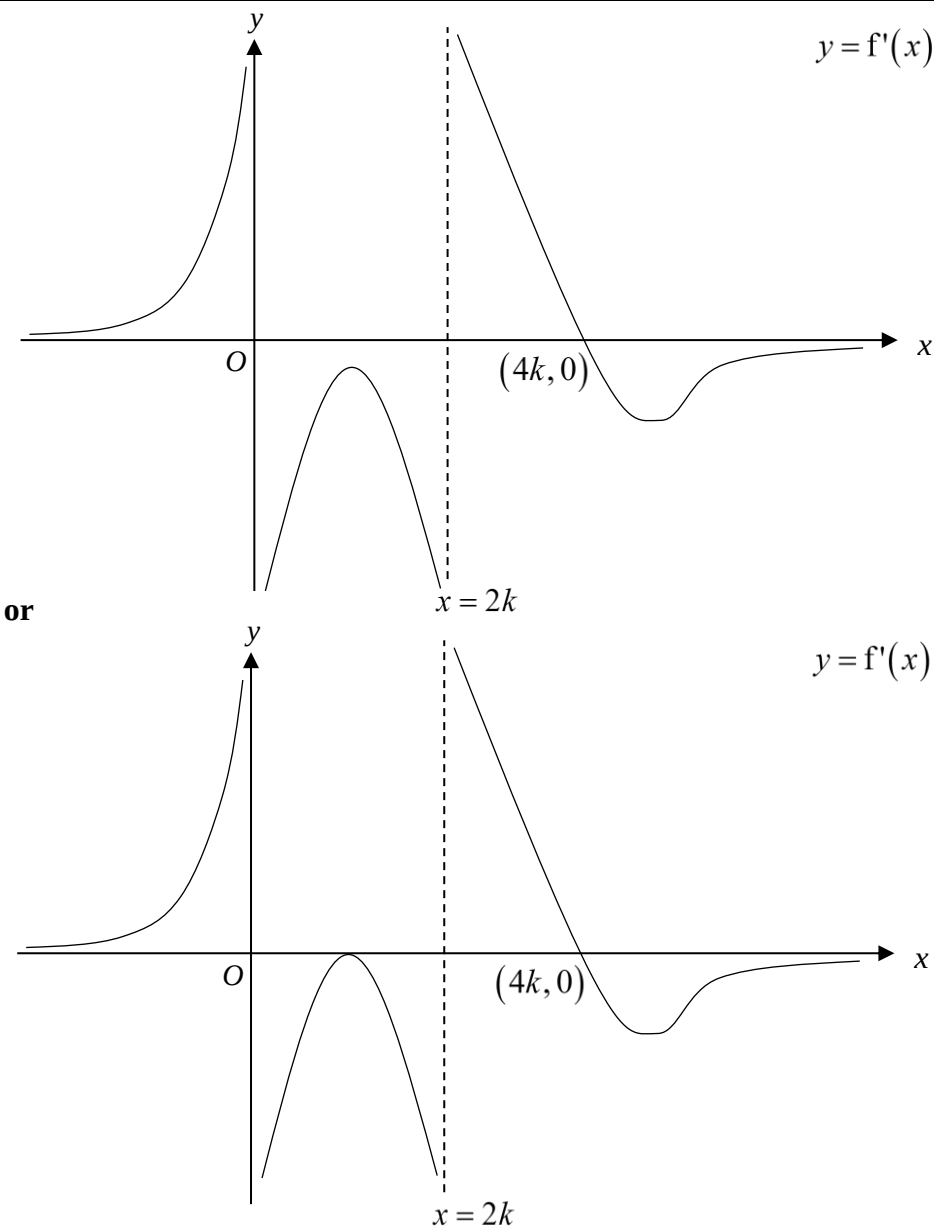
$$y = f(x - k)$$

↓ reflection in the y -axis

$$y = f(-x - k)$$

Relate the graph of $y = f'(x)$ to the graph of $y = f(x)$.

(ii)

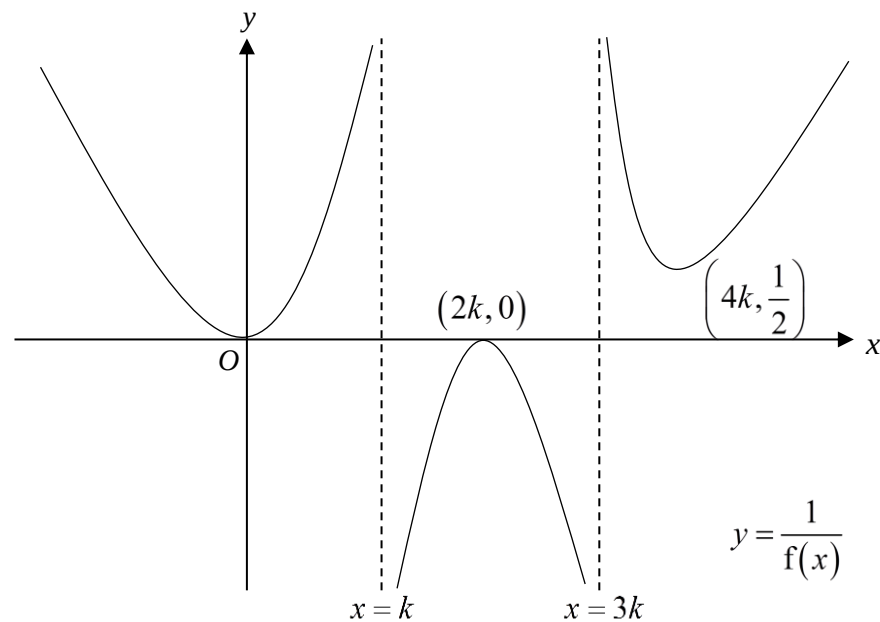


Many good answers. However, students should take note that they are supposed to state the **coordinates** of any turning point(s) and any points where the curve crosses the x- and y-axes as given in the question.

There were a number of students who mixed up the sketching of $y = f'(x)$ and $y = \frac{1}{f(x)}$.

Relate the graph of $y = \frac{1}{f(x)}$ to the graph of $y = f(x)$.

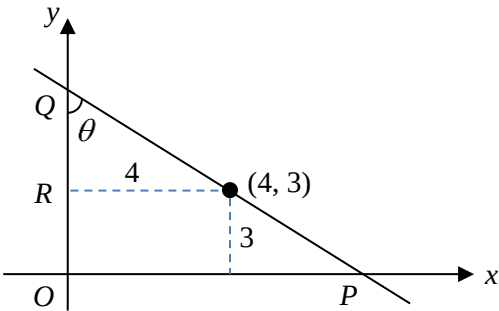
(iii)



Many good answers. However, students should take note that they are supposed to state the **coordinates** of any turning point(s) and any points where the curve crosses the x- and y-axes as given in the question.

There were a number of students who mixed up the sketching of $y = f'(x)$ and $y = \frac{1}{f(x)}$.

Q6. Application of Differentiation

Assessment Objectives	Solution	Examiner's Feedback
Use of trigonometric ratio to express gradient and y-intercept in terms of θ .	(i)  Gradient = $-\frac{1}{\frac{OP}{OQ}} = -\frac{1}{\tan \theta} = -\cot \theta$ $\tan \theta = \frac{4}{QR} \Rightarrow QR = 4 \cot \theta$ y-intercept = $3 + 4 \cot \theta$ Equation of line PQ is $y = -(\cot \theta)x + 3 + 4 \cot \theta$ $y = (4 - x) \cot \theta + 3$ (shown)	Poorly attempted. Many students could identify that they needed to find gradient but did not realized that gradient in this question is in fact negative. Students who attempted to 'work backwards' but did not show sufficient and accurate working were penalized.

Find axial intercepts using equation of line. Find stationary value using first derivative.	(ii) When $x = 0$, $y = 4 \cot \theta + 3$ When $y = 0$, $0 = (4 - x) \cot \theta + 3$ $x = 4 - \frac{-3}{\cot \theta} = 4 + 3 \tan \theta$ $OP + OQ = 4 + 3 \tan \theta + 4 \cot \theta + 3$ $= 7 + 3 \tan \theta + 4 \cot \theta$ Let $L = OP + OQ$ $\frac{dL}{d\theta} = 3 \sec^2 \theta - 4 \operatorname{cosec}^2 \theta$ $\frac{dL}{d\theta} = 0 \Rightarrow 3 \sec^2 \theta = 4 \operatorname{cosec}^2 \theta$ $\frac{3}{\cos^2 \theta} = \frac{4}{\sin^2 \theta}$ $\tan^2 \theta = \frac{4}{3}$ $\tan \theta = \frac{2}{\sqrt{3}} \left(\text{rej. } -\frac{2}{\sqrt{3}} \because 0 < \theta < \frac{\pi}{2} \right)$ Stationary value of $OP + OQ = 7 + 3 \left(\frac{2}{\sqrt{3}} \right) + 4 \left(\frac{\sqrt{3}}{2} \right)$ $= 7 + 4\sqrt{3}$	Many students were unable to find the x-coordinate of point P. For students who found the expression for $OP + OQ$, they were unable to differentiate the expression. Students should know the following: (1) $\frac{d}{d\theta}(\tan \theta) = \sec^2 \theta$ (2) $\frac{d}{d\theta}(\cot \theta) = -\operatorname{cosec}^2 \theta$
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Q7. Parametric Equations		
Assessment Objectives	Solution	Examiner's Feedback
Find first derivative of a function defined parametrically.	(i) $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2t}{-4(t+1)^2} = -\frac{t(t+1)^2}{2}$	Most students able to get the correct answers. Those who were unable to do made careless mistake in dx/dt.
Find equation of normal.	(ii) When $x = -2$, $\frac{4}{t+1} = -2$ $t = -3$ $y = 6$ Gradient of normal = $\frac{2}{-3(-3+1)^2} = -\frac{1}{6}$ Equation of normal at $P(-2, 6)$ is $y - 6 = -\frac{1}{6}(x + 2)$ $y = -\frac{1}{6}x + \frac{17}{3} \text{ or } 6y + x = 34$	Most students got the correct concept to solve for the eqn of normal but lost the accuracy mark due of the wrong expression in part (i).
Find t -values at points of intersection of a Cartesian line and a parametric curve.	(iii) $t^2 - 3 = -\frac{1}{6}\left(\frac{4}{t+1}\right) + \frac{17}{3}$ $6(t+1)(t^2 - 3) = -4 + 34(t+1)$ $3(t+1)(t^2 - 3) = 17t + 15$ $3t^3 + 3t^2 - 9t - 9 = 17t + 15$ $3t^3 + 3t^2 - 26t - 24 = 0$ Using GC, $t = -3$ (given) or $t = -0.915$ (3 s.f.) or $t = 2.91$ (3 s.f.)	Many students attempt to convert the parametric eqn of the curve to cartesian form first then solve for the x values, then solve for the t values which lead to a longer method. Please note that this method may not work for all questions as it may be hard/impossible to convert to cartesian form. Also, many students didn't make use of their GC to solve and hence wasted their time to solve algebraically.

Q8. Graphing Techniques		
Assessment Objectives	Solution	Examiner's Feedback
Perform long division	(i) $y = \frac{2x^2 - 3x + 5}{x - 5} = 2x + 7 + \frac{40}{x - 5}$ $\therefore p = 2$ $q = 7$ $r = 40$ $\begin{array}{r} 2x+7 \\ x-5 \overline{) 2x^2-3x+5} \\ \underline{-(2x^2-10x)} \\ 7x+5 \\ \underline{-(7x-35)} \\ 40 \end{array}$	This part was generally well done.
Identify characteristic of asymptotes, turning points and axial intercepts. Use of a G.C. to graph a given function.	(ii) Asymptotes: $y = 2x + 7$ and $x = 5$ $y = \frac{2x^2 - 3x + 5}{x - 5}$ The graph shows the function $y = \frac{2x^2 - 3x + 5}{x - 5}$ plotted on a Cartesian coordinate system. The vertical asymptote is a dashed line at $x = 5$. The slant asymptote is a dashed line at $y = 2x + 7$. The graph consists of two branches. The upper branch has a minimum point at $(9.47, 34.9)$. The lower branch passes through the y-intercept at $(0, -1)$ and the x-intercept at $(0.528, -0.889)$. The origin is labeled O.	Candidates need to use a ruler for axis and asymptotes to provide an accurate sketch. Many students did not write down intercept in coordinate form.

Understand that the number of intersections is equivalent to the number of roots in an equation.	(ii) $(2x^2 - 3x + 5)^2 = 5x(x - 5)^2$ $\left(\frac{2x^2 - 3x + 5}{x - 5}\right)^2 = 5x$ $y^2 = 5x$ $y = \pm\sqrt{5x}$ Sketch $y = \pm\sqrt{5x}$ in part (ii). From the diagram, there are 2 points of intersections. Hence, there are 2 roots.	Many students did not include $y = -\sqrt{5x}$. Some students drew a sketch of $y = \pm\sqrt{5x}$ which did not touch the origin.
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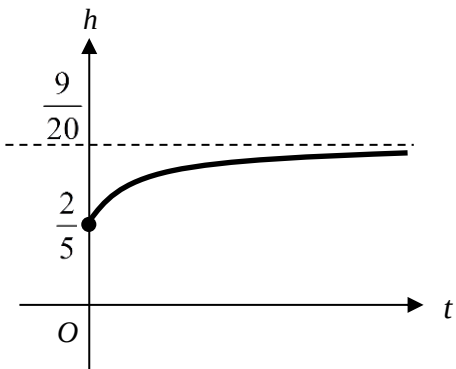
Q9. Maclaurin's Series		
Assessment Objectives	Solution	Examiner's Feedback
Use of series expansion formula in MF26.	(a) Method ①: $\sqrt{4-x} = 4^{\frac{1}{2}} \left(1 - \frac{x}{4}\right)^{\frac{1}{2}}$ $= 2 \left(1 - \frac{x}{8} + \dots\right)$ $= 2 - \frac{x}{4} + \dots$ $p + \ln(q-x) = p + \ln \left[(q) \left(1 - \frac{x}{q}\right) \right]$ $= p + \ln q + \ln \left(1 - \frac{x}{q}\right)$ $= (p + \ln q) - \frac{x}{q} + \dots$ Comparing, $2 = p + \ln q \quad \text{and} \quad -\frac{x}{4} = -\frac{x}{q}$ $2 = p + \ln 4 \quad q = 4$ $p = 2 - \ln 4$ Method ②: Let $f(x) = \sqrt{4-x} \Rightarrow f'(x) = \frac{-1}{2\sqrt{4-x}}, \therefore f(0) = 2 \text{ \& } f'(0) = -\frac{1}{4}$ Let $g(x) = p + \ln(q-x) \Rightarrow g'(x) = \frac{-1}{q-x}, \therefore g(0) = p + \ln q \text{ \& } f'(0) = -\frac{1}{q}$ Comparing, $q = 4$ and $p = 2 - \ln 4$	Quite a majority of the students attempted this question successfully with a variety of methods. The most successful method being the use of repeated derivatives to form equations in p and q. Common errors include erroneous use of the standard series expansions and also not knowing how to convert the expressions into the standard form required in their use. A significant number of students also made arithmetic errors on the rules of logarithms, resulting in many marks lost.

Implicit differentiation involving trigonometric expressions. Use of formula given in MF26 to find Maclaurin series.	(i) $y = \tan^{-1}(ax+1)$ $\tan y = ax+1$ $\sec^2 y \frac{dy}{dx} = a$ $\frac{dy}{dx} = a \cos^2 y$ (shown) $\frac{d^2 y}{dx^2} = 2a \cos y (-\sin y) \frac{dy}{dx} = -a \sin 2y \frac{dy}{dx}$ $\frac{d^3 y}{dx^3} = -2a \cos 2y \left(\frac{dy}{dx}\right)^2 - a \sin 2y \frac{d^2 y}{dx^2}$ When $x = 0$, $y = \tan^{-1}(1) = \frac{\pi}{4}$ $\frac{dy}{dx} = a \left(\cos \frac{\pi}{4}\right)^2 = \frac{1}{2}a$ $\frac{d^2 y}{dx^2} = -a \left(\sin \frac{\pi}{2}\right) \left(\frac{1}{2}a\right) = -\frac{1}{2}a^2$ $\frac{d^3 y}{dx^3} = -2a \left(\cos \frac{\pi}{2}\right) \left(\frac{1}{2}a\right)^2 - a \left(\sin \frac{\pi}{2}\right) \left(-\frac{1}{2}a^2\right) = \frac{1}{2}a^3$ $\tan^{-1}(ax+1) = \frac{\pi}{4} + \frac{1}{2}ax + \frac{-\frac{1}{2}a^2}{2!}x^2 + \frac{\frac{1}{2}a^3}{3!}x^3 + \dots$ $= \frac{\pi}{4} + \frac{1}{2}ax - \frac{1}{4}a^2x^2 + \frac{1}{12}a^3x^3 + \dots$	Most students performed badly for this question as they are unclear about the process of implicit differentiation, often omitting the multiplication of the first derivative. Students who attempted direct differentiation are rarely successful due to the complexity of the equations. Most students who are successful with the repeated differentiation ended up with the correct expression, except a few who made arithmetic errors on the coefficients.
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Use of chain rule and formula given in MF26 to differentiate $\tan^{-1}(4x+1)$, and make use of expression found in (i).	(ii) Method ① (HENCE: direct differentiation using MF26) $\frac{d}{dx}[\tan^{-1}(4x+1)] = \frac{4}{1+(4x+1)^2}$ $\frac{1}{1+(4x+1)^2} = \frac{1}{4} \frac{d}{dx}[\tan^{-1}(4x+1)]$ $= \frac{1}{4} \frac{d}{dx}\left[\frac{\pi}{4} + 2x - 4x^2 + \frac{16}{3}x^3 + \dots\right]$ $= \frac{1}{4}[2 - 8x + 16x^2 + \dots]$ $= \frac{1}{2} - 2x + 4x^2 + \dots$ Method ② (OTHERWISE: binomial expansion) $\frac{1}{1+(4x+1)^2} = [1 + (16x^2 + 8x + 1)]^{-1}$ $= 2^{-1}[1 + (4x + 8x^2)]^{-1}$ $= \frac{1}{2}\left[1 - (4x + 8x^2) + \frac{(-1)(-2)}{2!}(4x)^2 + \dots\right]$ $= \frac{1}{2}[1 - 4x - 8x^2 + 16x^2 + \dots]$ $= \frac{1}{2} - 2x + 4x^2 + \dots$ Method ③ (OTHERWISE: repeated differentiation) $f(x) = \frac{1}{1+(4x+1)^2} \Rightarrow f'(x) = \frac{-8(4x+1)}{[1+(4x+1)^2]^2} \Rightarrow f(0) = \frac{1}{2} \text{ \& } f'(0) = -2$ $\Rightarrow f''(x) = \frac{-32[1+(4x+1)^2]^2 + 128(4x+1)^2[1+(4x+1)^2]}{[1+(4x+1)^2]^4} \Rightarrow f''(0) = 8$ $\therefore f(x) = \frac{1}{2} - 2x + 4x^2 + \dots$	Most students were unable to see the link necessary for the “hence” method and adopted the otherwise methods. The most successful methods were those which involved repeated differentiation as it does not depend on the previous answers. Many students attempted to use the series expansion for $(1+x)^n$ using $(4x+1)^2$ in place of x, but failing to realize that all powers of $(4x+1)$ will result in terms which have to be include (i.e. constant, x and x^2).
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Q10. Differential Equations		
Assessment Objectives	Solution	Examiner's Feedback
Use of chain rule Formulate differential equation from a problem situation	(i) $V = \pi(4^2)h$ $\frac{dV}{dh} = 16\pi$ $\frac{dV}{dt} = \frac{dV_{in}}{dt} - \frac{dV_{out}}{dt}$ $= 0.36\pi - 0.8\pi h$ $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ $0.36\pi - 0.8\pi h = 16\pi \cdot \frac{dh}{dt}$ $\frac{dh}{dt} = \frac{0.36\pi - 0.8\pi h}{16\pi}$ $\frac{dh}{dt} = \frac{1}{400}(9 - 20h) \text{ (shown)}$	This part is usually well done. Some candidates introduced t, representing time, in an attempt to establish an equation of h in terms of t. Followed by wrong differentiation of h with respect to t. This approach earn no mark.

Solve differential equations to find particular solution.	(ii) $\frac{dh}{dt} = \frac{1}{400}(9 - 20h)$ $\int \frac{1}{9 - 20h} dh = \frac{1}{400} \int 1 dt$ $-\frac{1}{20} \int \frac{-20}{9 - 20h} dh = \frac{1}{400} \int 1 dt$ $-\frac{1}{20} \ln 9 - 20h = \frac{1}{400}(t + A)$ $\ln 9 - 20h = -\frac{1}{20}(t + A)$ $ 9 - 20h = e^{-\frac{1}{20}(t+A)}$ $9 - 20h = \pm e^{-\frac{1}{20}(t+A)}$ $9 - 20h = \pm e^{-\frac{1}{20}t} \cdot e^{-\frac{1}{20}A}$ $9 - 20h = Be^{-\frac{1}{20}t} \text{ where } B = \pm e^{-\frac{1}{20}A}$ When $t = 0, h = 0.4,$ $9 - 20(0.4) = Be^{-\frac{1}{20}(0)}$ $B = 1$ $\therefore 9 - 20h = e^{-\frac{1}{20}t}$ $20h = 9 - e^{-\frac{1}{20}t}$ $h = \frac{1}{20} \left(9 - e^{-\frac{1}{20}t} \right)$	Careless mistakes in writing the numbers are unusually frequent in this part and resulted in marks loss. Generally well done.
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Interpret a differential equation and its solution in terms of a problem situation.	(iii) If the tap is on indefinitely, the tank will not be empty. In the long run, there will be $\frac{9}{20}$ m of water in the tank.	No marks awarded to candidates who attempted to explain in words without clear reference to the mathematical equation obtained earlier.
Interpret a differential equation and sketch a graph.	(iv) $h = \frac{1}{20} \left(9 - e^{-\frac{1}{20}t} \right)$ 	Many candidates often overlooked the presence of a horizontal asymptote. Lacks proper labelling of axes or asymptote.

Q11. Vectors

Assessment Objectives	Solution	Examiner's Feedback
Solve a two-dimensional vector geometry problem involving abstract vectors, by : - Using the collinearity theorem to formulate expressions for the position vector of an unknown point, in terms of two non-zero non-parallel base vectors, and - Compare and equate corresponding coefficients of respective base vectors in a vector equation, to solve the problem.	(i) $\overrightarrow{AM} = -\mathbf{a} + \frac{1}{2}\mathbf{b}$ $\overrightarrow{BN} = -\mathbf{b} + \frac{1}{2}\mathbf{a}$ $\overrightarrow{OX} = \mathbf{a} + \overrightarrow{AX} = \mathbf{b} + \overrightarrow{BX}$ $= \mathbf{a} + \lambda(-\mathbf{a} + \frac{1}{2}\mathbf{b}) = \mathbf{b} + \mu(-\mathbf{b} + \frac{1}{2}\mathbf{a})$ for some scalars λ, μ $(1-\lambda)\mathbf{a} + \frac{\lambda}{2}\mathbf{b} = \frac{\mu}{2}\mathbf{a} + (1-\mu)\mathbf{b}$ $\therefore \begin{cases} 1-\lambda = \frac{\mu}{2} \\ \frac{\lambda}{2} = 1-\mu \end{cases}$ Solving, $\lambda = \mu = \frac{2}{3}$ $\therefore \overrightarrow{OX} = \mathbf{a} + \frac{2}{3}(-\mathbf{a} + \frac{1}{2}\mathbf{b}) = \frac{1}{3}\mathbf{a} + \frac{1}{3}\mathbf{b}$ (shown) Method ②: (Using Equation of Lines) $l_{AM} : \mathbf{r} = \mathbf{a} + \lambda\left(-\mathbf{a} + \frac{1}{2}\mathbf{b}\right), \lambda \in \mathbb{R}$ $l_{BN} : \mathbf{r} = \mathbf{b} + \mu\left(-\mathbf{b} + \frac{1}{2}\mathbf{a}\right), \mu \in \mathbb{R}$ Since X lies on both lines, $\mathbf{a} + \lambda\left(-\mathbf{a} + \frac{1}{2}\mathbf{b}\right) = \mathbf{b} + \mu\left(-\mathbf{b} + \frac{1}{2}\mathbf{a}\right)$ $\begin{cases} 1-\lambda = \frac{\mu}{2} \\ \frac{\lambda}{2} = 1-\mu \end{cases}$ Solving, $\lambda = \mu = \frac{2}{3}$ $\therefore \overrightarrow{OX} = \mathbf{a} + \frac{2}{3}(-\mathbf{a} + \frac{1}{2}\mathbf{b}) = \frac{1}{3}\mathbf{a} + \frac{1}{3}\mathbf{b}$ (shown)	Most students were able to obtain at least 2 out of 4 marks by using ratio theorem to find $\overrightarrow{AM}$ and $\overrightarrow{BN}$, but some were unsure how to continue. Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel, we can compare the coefficients of the 2 vectors to obtain the 2 equations.

	Method ②: (Using Ratio Theorem) Using triangle OAM, $\overrightarrow{OX} = \frac{\lambda(\mathbf{a}) + (1-\lambda)\left(\frac{1}{2}\mathbf{b}\right)}{\lambda + (1-\lambda)} = \lambda\mathbf{a} + \frac{(1-\lambda)}{2}\mathbf{b}$ Using triangle ONB, $\overrightarrow{OX} = \frac{\mu\left(\frac{1}{2}\mathbf{a}\right) + (1-\mu)\mathbf{b}}{\mu + (1-\mu)} = \frac{1}{2}\mu\mathbf{a} + (1-\mu)\mathbf{b}$ $\lambda\mathbf{a} + \frac{(1-\lambda)}{2}\mathbf{b} = \frac{1}{2}\mu\mathbf{a} + (1-\mu)\mathbf{b}$ $\begin{cases} \lambda = \frac{\mu}{2} \\ \frac{1-\lambda}{2} = 1-\mu \end{cases}$ Solving, $\lambda = \frac{1}{3}$, $\mu = \frac{2}{3}$ $\overrightarrow{OX} = \frac{1}{3}\mathbf{a} + \frac{1}{3}\mathbf{b} \quad (\text{shown})$	
Apply the midpoint theorem. Apply the collinearity theorem to determine whether three distinct points are collinear.	(ii) $\overrightarrow{OT} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$ using the midpoint theorem $\overrightarrow{OX} = \frac{1}{3}(\mathbf{a} + \mathbf{b})$ $= \frac{2}{3}\left[\frac{1}{2}(\mathbf{a} + \mathbf{b})\right] = \frac{2}{3}\overrightarrow{OT}$ Since $\overrightarrow{OX} = k\overrightarrow{OT}$ for some scalar k where $0 < k < 1$, $\overrightarrow{OX}$ is parallel to $\overrightarrow{OT}$ with a common point O, hence X lies on OT.	Most students gave an incomplete proof for X lying on OT. It is essential to show that $\overrightarrow{OX}$ is a scalar multiple of $\overrightarrow{OT}$ and hence the 2 vectors are parallel.

Find a normal vector for a plane given three non-collinear points on the plane.

Formulate a vector equation of a plane in scalar product form, using a point on the plane and a normal vector to the plane.

(iii)

$$\overrightarrow{OA} = \begin{pmatrix} 5 \\ 4 \\ 6 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix}$$

$$\begin{aligned} \overrightarrow{OA} \times \overrightarrow{OB} &= \begin{pmatrix} 5 \\ 4 \\ 6 \end{pmatrix} \times \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} (4)(3) - (6)(2) \\ (6)(-2) - (5)(3) \\ (5)(2) - (4)(-2) \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ -27 \\ 18 \end{pmatrix} = 9 \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} \end{aligned}$$

Since $\begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix}$ is perpendicular to the plane, and origin O is on the plane,

$$\text{it is represented by } \mathbf{r} \cdot \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} = 0.$$

$$\therefore -3y + 2z = 0 \quad (\text{shown})$$

Most students were able to obtain the normal of the plane.

Since O is on the plane, the most direct method is to cross $\overrightarrow{OA}$ and $\overrightarrow{OB}$.

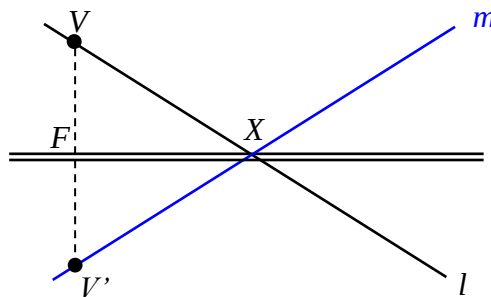
Find the foot of the perpendicular from a given point to a given plane, by: - Formulate an equation for the perpendicular line passing through the point, and - Find the point of intersection between this perpendicular line and the plane.	(iv) Line VF, $l_{VF} : \mathbf{r} = \begin{pmatrix} 1 \\ -68 \\ -37 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix}, \lambda \in \mathbb{R}.$ Since F is on l_{VF}, $\overrightarrow{OF} = \begin{pmatrix} 1 \\ -68 \\ -37 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix}$, for some $\lambda \in \mathbb{R}.$ Since F is on p, $\overrightarrow{OF} \cdot \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} = 0.$ $\Rightarrow \left[\begin{pmatrix} 1 \\ -68 \\ -37 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} \right] \cdot \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} = 0$ $130 + 13\lambda = 0$ $\lambda = -10$ $\overrightarrow{OF} = \begin{pmatrix} 1 \\ -68 \\ -37 \end{pmatrix} + (-10) \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -38 \\ -57 \end{pmatrix}$ The coordinates of F is $(1, -38, -57).$	Some students had the misconception that $\overrightarrow{VF} = \begin{pmatrix} 1 \\ -68 \\ -37 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} \text{ and that since } \overrightarrow{VF} \text{ is parallel to the normal of plane, } \overrightarrow{VF} \cdot \begin{pmatrix} 0 \\ -3 \\ 2 \end{pmatrix} = 0. \text{ The 2 vectors are parallel, not perpendicular.}$ A number of students were careless in solving for the value of $\lambda.$
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Given a line and a plane that intersects at a point, construct a vector equation for the reflection of a line in a plane, by :

- Locating the point of intersection between the line and the plane,
- Finding the point of reflection of another point on the line in the plane, and
- Constructing a vector equation of the reflected line containing these two points.

Convert a vector equation for the line into a Cartesian equation.

(v)
$$\overrightarrow{OX} = \frac{1}{3}(\overrightarrow{OA} + \overrightarrow{OB}) = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$



Let V' be the reflection of V in plane p .

$$\overrightarrow{OF} = \frac{\overrightarrow{OV} + \overrightarrow{OV'}}{2} \quad \left[\text{or use } \overrightarrow{VF} = \overrightarrow{FV'} \right]$$

$$\overrightarrow{OV'} = 2\overrightarrow{OF} - \overrightarrow{OV}$$

$$\overrightarrow{OV'} = 2 \begin{pmatrix} 1 \\ -38 \\ -57 \end{pmatrix} - \begin{pmatrix} 1 \\ -68 \\ -37 \end{pmatrix} = \begin{pmatrix} 1 \\ -8 \\ -77 \end{pmatrix}$$

$$\overrightarrow{VX} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 1 \\ -8 \\ -77 \end{pmatrix} = \begin{pmatrix} 0 \\ 10 \\ 80 \end{pmatrix} = 10 \begin{pmatrix} 0 \\ 1 \\ 8 \end{pmatrix}$$

$$\text{Line } m: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + k \begin{pmatrix} 0 \\ 1 \\ 8 \end{pmatrix}, k \in \mathbb{R}$$

$$\text{Line } m: x=1, y-2 = \frac{z-3}{8}$$

Some students failed to notice that X is the centroid of triangle OAB although it was mentioned in the question.

Common mistakes include using $\overrightarrow{OX} = \frac{\overrightarrow{OV} + \overrightarrow{OV'}}{2}$ instead of $\overrightarrow{OF}$.

The above mistake could have been avoided if the student had **drawn a diagram**.

The question asked for a cartesian equation of line m , hence students were penalized for giving the vector equation form as the final answer.